Q.30 Differentiate between direct and indirect methods for measurement. (CO1)

Q.31 Explain the working of deflection type instrument. (CO2)

Q.32 List the errors in dynamometer type wattmeter. (CO2)

Q.33 Draw and explain the block diagram of basic measurement system. (CO1)

Q.34 Write a short note on megger. (CO6)

Q.35 Explain power factor and phase sequence of a three phase system. (CO6)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Give the working of (CO4)

a) Current transformer b) Potential transformer

Q.37 Explain measurement of three phase power for 3 wire and 4 wire system with diagram for star connected load. (CO2)

Q.38 Explain the construction and working of Energy meter with diagram. (CO2)

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Roll No. ....

4th Sem / Elect, Power Station Engg, Elect. & Eltx. Engg

Subject:-Electrical Measuring Instruments and Instrumentations/ Elect. & Eltx. Measuring Instr.

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Energy meter is a type of_______. (CO2)

a) Indicating meter c) Integrating meter

b) Recording meter d) none of these

Q.2 Indicating instruments should be (CO1)

a) Undamped c) Critically damped

b) Underdamped d) Overdamped

Q.3 A moving iron instrument can be used for (CO2)

a) Only AC c) Only DC

b) Both ac or dc d) high frequency ac

Q.4 To extend the range of an ammeter, a resistance is connected in (CO3)

a) parallel c) series

b) parallel-series d) no resistance required

Q.5 A dynamometer wattmeter has (CO2)

a) Pressure coil as moving coil

b) current coil as moving coil

c) any of the (a) and (b) as per requirement

d) none of these

Q.6 Power is defined as ________ (CO2)

a) Voltage X current c) Voltage X time

b) Voltage / current d) Voltage / time

Q.7 Extremely low resistance can be measured by (CO6)

a) Megger c) Kelvin bridge

b) ohmmeter d) Earth tester

Q.8 O in CRO stands for (CO6)

a) Oscillator c) omniscope

b) Oxciloscope d) oscilloscope

Q.9 A phase sequence indicator works on the principle of (CO6)

a) Electromagnetic induction

b) Electrodynamic effect

c) Electrostatic effect

d) Electro thermal effect

Q.10 Instrument transformer cannot be used to measure (CO4)

a) DC current c) DC power

b) DC voltage d) all the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 A voltmeter is always connected in series with the circuit. (True/False) (CO2)

Q.12 A moving coil ammeter measures ____ ___ current. (CO2)

Q.13 Maximum demand indicator works on the principle of_______. (CO6)

Q.14 Give applications of synchroscope. (CO6)

Q.15 Name methods used to measure power in 3 phase 3 wire system. (CO2)

Q.16 Name any recording type instrument. (CO1)

Q.17 Give applications of gravity control method. (CO1)

Q.18 Why an ammeter has low resistance? (CO1)

Q.19 What is the use of earth tester? (CO6)

Q.20 On which phenomenon thermocouple works? (CO7)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain the working principle of LVDT. (CO7)

Q.22 Explain deflecting, controlling and damping toque in brief. (CO2)

Q.23 Write a short note on phase difference measurement using CRO. (CO7)

Q.24 State applications where multimeter is used. (CO2)

Q.25 List the advantages and disadvantages of LCR meter. (CO6)

Q.26 Explain the working of Permanent magnet moving coil instrument. (CO2)

Q.27 What do you mean by active and passive transducer with example? (CO7)

Q.28 Write a short note on Maximum demand indicators. (CO6)

Q.29 Explain the working principle of repulsion type moving instrument. (CO2)